

IKU Department of Computer Engineering

Course Outline

SEN0414_Advanced Programming (2+2 - 3)

Instructor: **Yusuf Altunel**

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Content

1. **Part I: Python Fundamentals**
 - a. Chapter 1: Python Basics
 - b. Chapter 2: if-else and Flow Control
 - c. Chapter 3: Loops
 - d. Chapter 4: Functions
 - e. Chapter 5: Debugging
 - f. Chapter 6: Lists
 - g. Chapter 7: Dictionaries and Structuring Data
2. **Part II: Text, Files and the Filesystem**
 - a. Chapter 8: Strings and Text Editing
 - b. Chapter 9: Text Pattern Matching with Regular Expressions
 - c. Chapter 10: Reading and Writing Files
 - d. Chapter 11: Organizing Files
 - e. Chapter 12: Designing and Deploying Command Line Programs
3. **Part III: The Web, Spreadsheets and Databases**
4. **Part IV: Automating the Desktop and Beyond**
 - a. Chapter 13: Web Scraping
 - b. Chapter 14: Excel Spreadsheets
 - c. Chapter 15: Google Sheets
 - d. Chapter 16: SQLite Databases
 - e. Chapter 17: PDF and Word Documents
 - f. Chapter 18: CSV, JSON, and XML Files
- a. Chapter 19: Keeping Time, Scheduling Tasks, and Launching Programs
- b. Chapter 20: Sending Email, Texts, and Push Notifications
- c. Chapter 21: Making Graphs and Manipulating Images
- d. Chapter 22: Recognizing Text in Images
- e. Chapter 23: Controlling the Keyboard and Mouse
- f. Chapter 24: Text-to-Speech and Speech Recognition Engines

Grading

Grading Item	BONUS	Midterm	Project	Final	Totals
%	10	30	30	40	110

(10 Points bonus!)

Notice: This term theme is semantic technologies, so you are expected to develop industrial or educational applications for real usages, not just hypothetical. So, you need to communicate with the real users, properly experiment domain engineering facilities and collect business requirements. Team projects with well-defined responsibilities are in favour.

Textbook

[1] Automate the Boring Stuff with Python, 3e; Al **Sweigart**; No Starch Press, April 2025, 672 pp., ISBN-13: 9781718503403. Free to read online at <https://automatetheboringstuff.com/3e/>

Other references

[2] Python Crash Course, 3e; Eric **Matthes**; No Starch Press, December 2022, 552 pp., ISBN-13: 9781718502703

[3] Programming Python, 4e; Mark **Lutz**; O'Reilly, 2011 (no later edition published).

[4] Introduction to Python; Mark **Lassoff** & Julius **Hernandez**; LearnToProgram.tv, Incorporated, 2018

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- [5] The Python Tutorial and The Python Standard Library, Python Software Foundation, version 3.14.7. Free: <https://docs.python.org/3/tutorial/> and <https://docs.python.org/3/library/>
- [6] PEP 8 – Style Guide for Python Code; Guido van Rossum, Barry Warsaw, Alyssa Coghlan; Python Software Foundation. Free: <https://peps.python.org/pep-0008/>
- [7] Think Python, 3e; Allen B. Downey; Green Tea Press. Free under a Creative Commons licence: <https://alldowney.github.io/ThinkPython/>
- [8] Python for Everybody: Exploring Data in Python 3; Charles R. Severance. Free under Creative Commons Attribution 4.0: <https://www.py4e.com/book>
- [9] The Big Book of Small Python Projects; Al Sweigart; No Starch Press, June 2021, 432 pp., ISBN-13: 9781718501249. Free to read: <https://inventwithpython.com/bigbookpython/>

Lecture	Content	Date	Hours	Venue
Hour-1	Theory & Practices	Friday	14:00-14:45	2B PC Lab 12/14
Hour-2			15:00-15:45	
Hour-3			16:00-16:45	
Hour-4			17:00-17:45	

Academic Integrity: The Computer Engineering Department of IKU draws your attention to the concepts of “Academic Integrity”. You should obey to written and verbal rules of studentship, otherwise you will be reported to the faculty and lose all bonus fortunes of this course.